

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
 Semester: July 2019 - November 2019

Max. Marks: 100

Duration: 3 Hrs

End Semester Exam

Class: S.Y.BTech Semester: III

Name of the Course: Integral Transform and Vector Calculus

Branch: COMP

Course Code: 2UCC301

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question No.		Max. Marks
Q.1	A Attempt the following	
	Find $L\{t^5 + \sinh(2t)\}$	05
	B Attempt any THREE .	
	(a) Define Laplace transform of $f(t)$ and hence evaluate $\int_0^\infty \frac{\sin 3t + \sin 2t}{te^t} dt$	
	(b) Find Inverse Laplace Transform of $\left\{s \log \left(\frac{s+1}{s-1}\right)\right\}$	
	(c) Express the following function as Heaviside Unit Step function and find its Laplace transform $f(t) = \begin{cases} \cos t, & 0 < t < \frac{\pi}{2} \\ \sin t, & t > \frac{\pi}{2} \end{cases}$	21
	(d) Using Laplace Transform, solve the following differential equation $(D^2 + D)y = t^2 + 2t, y(0) = 4, y'(0) = -2$	
Q.2	A Attempt the following	
	Find Fourier series of $f(x) = x \cos(x), -\pi \leq x \leq \pi$	05
	B Attempt any THREE .	
	(a) Find the Fourier series for $f(x) = x \sin x$ in the interval $0 \leq x \leq 2\pi$.	
	(b) Obtain Complex form of Fourier series for e^{2x} in $(-2, 2)$	
	(c) Find half range cosine series for $f(x) = x$ in $(0, 2)$ and hence deduce that $\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$	21
	(d) Find Fourier Transform of $f(x) = \begin{cases} 1 - x , & x < 1 \\ 0, & x > 1 \end{cases}$ and hence show that $\int_0^\infty \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$	
	(e) Find Fourier Integral for $f(x) = \begin{cases} 1 - x^2 & \text{for } x \leq 1 \\ 0 & \text{for } x > 1 \end{cases}$	
Q.3	Attempt any TWO	
	(a) Find Z-Transform and its ROC of $f(k) = 2^k \cos k, k \geq 0$	
	(b) Find Z-Transform and its ROC of $f(k) = k^3$ using $Z\{1\}, k \geq 0$	10
	(c) Find Inverse Z-Transform for $\frac{1}{z-3}, z < 3, z > 3$	

Q.4	A	Attempt the following	04
		Prove that $i \times (\bar{a} \times i) + j \times (\bar{a} \times j) + k \times (\bar{a} \times k) = 2\bar{a}$	
	B	Attempt any TWO.	14
	(a)	Show that $\bar{F} = (y^2 - z^2 + 3yz - 2x)i + (3xz + 2xy)j + (3xy - 2xz + 2z)k$ is both solenoidal & irrotational.	
	(b)	Prove that $\nabla \cdot \left\{ r \nabla \frac{1}{r^3} \right\} = \frac{3}{r^4}$	
Q.5	(c)	Find the constants a and b so that the surface $ax^2 - byz = (a + 2)x$ is orthogonal to the surface $4x^2y + z^3 = 4$ at $(1, -1, 2)$.	06
	A	Attempt the following	
		If $\bar{F} = (6xy + z^3)i + (3x^2 - z)j + (3xz^2 - y)k$ be the force field, find the work done in moving a particle from $(1, 2, -4)$ to $(3, 3, 2)$.	14
	B	Attempt any TWO .	
	(a)	Verify Green's Theorem in the plane for $\int_C (x^2 - y)dx + (2y^2 + x)dy$ around the boundary of the region defined by $y = x^2$ and $y = 4$.	
	(b)	Use Stokes theorem to evaluate $\int_C \bar{F} \cdot d\bar{r}$ where $\bar{F} = (2x - y)i - yz^2j - y^2zk$ and S is the surface of the hemisphere $x^2 + y^2 + z^2 = a^2$ lying above the xy -plane.	
	(c)	Use Gauss Divergence theorem to evaluate $\iint_S \bar{N} \cdot \bar{F} ds$ where $\bar{F} = 4xi - 2y^2j + z^2k$ and S is the region bounded by $x^2 + y^2 = 4, z = 0, z = 3$.	